PRIME MINISTER'S S OFFICE,
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT
KOROGWE DISTRICT COUNCIL

MOMBO DIVISION JOINT EXAMINATION SYNDICATE
FORM FOUR BEST TEN STUDENTS' EXAMINATION
041 BASIC MATHEMATICS
TIME: 3 HOURS **MONDAY 30TH MARCH 2026.**

INSTRUCTIONS:

wazaelimu.com

1. This paper consists of **Sections A** and **B.**
2. Answer **ALL** questions in each section. Each question in section A carries six (6) marks while each question in section B carries ten (10) marks.
3. All necessary working and answers for each question done must be shown clearly.
4. Mathematical tables may be used.
5. Non programmable Calculators may be used while Cellular phones are not allowed in the Examination room.
6. Write your **Examination number** on every page of your answer booklet(s).
7. The following constants may be used: The Radius of the Earth R = 6400km and π= 3.14

SECTION A (60 marks)
Answer ALL questions from this section

1. (a) A shopkeeper sold 392 T-shirts at the price of Tshs.7,950/= each. Approximate how much money he got.

   (b) Express $0.9\dot{6}$ in the form of $\frac{a}{b}$ where a and b are integers such that b≠0.

2. (a) Solve the value of x if $\left(\frac{1}{16}\right)^{x+3} \times \left(\frac{1}{32}\right)^{-x} = 1$

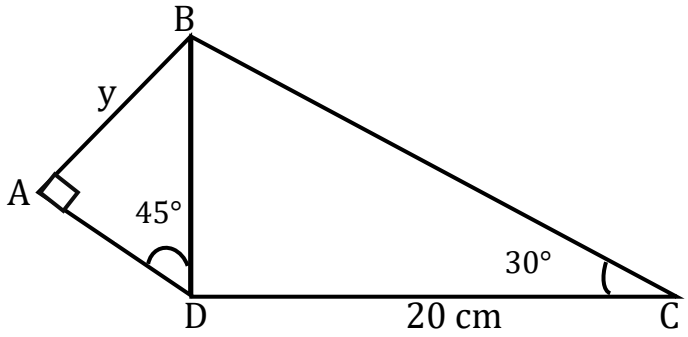
   (b) i) Simplify: $\sqrt{24} - \sqrt[3]{8} + \sqrt[2]{18}$

   ii) Given that log 2 = 0.3010, log 5 = 0.6990, log 7 = 0.8451 Evaluate $\log\left(\frac{35}{2}\right)$

3. (a) There are 48 men at a meeting of whom 24 are teachers, 36 are parents and 16 are both teachers and parents. By using Venn diagram, find the number of men who are neither teachers nor parents.

(b) A bag contains 6 white balls and 3 yellow balls. A ball is selected at random and not replaced, another ball is then selected. Find the probability of selecting one white ball and one yellow ball.

4. (a) If $a = 2i + 3j$, $b = -9j$ and $C = -3i - 2j$ find the magnitude of $a - b + c$
   (b) Find the equation of the line through the point (5, 3) perpendicular to the line whose equation is $3x - 4y = -1$
5. (a) The length of two sides of a triangle are 16 cm and 20 cm. Find the area of the triangle if the inclined angle is 30°.
   (b) Calculate the size of an interior angle of a regular nonagon.
6. (a) The price of TV set which includes V.A.T is Tshs.140,800/=. If the rate of V.A.T. is 30%, find the price of the TV before V.A.T was added.
   (b) 9 workers working 8 hours a day to complete a piece of work in 52 days. How long will it take 13 workers to complete the same job by working 6 hours a day.
7. (a) The ratio of men: women: children living in Momela village is 6:7:3. If there are 42,000 women, how many
   i. Children live in Momela village
   ii. People altogether live in Momela village.
   (b) If the 42000 women is an increase of 20% of the number of women 10 years ago. How many women lived in the village 10 years ago?
8. The third, fourth and eighth terms of an A.P forms the first three consecutive term of a G.P. If the sum of the first 10 terms of the A.P is 85, find:
   i. The first term of the arithmetic progression
   ii. The common ratio of G.P.
   iii. The sum of the first 7 terms of the G.P.
9. (a) Find the length marked y in four significant figures.



(b) A 4m ladder rests against a vertical wall with its foot 2m from the wall. How far up the wall does the ladder reached? (Give your answer in two decimal places).

10. (a) Factorize completely $2x^2 + x - 10$ by splitting the middle term.
    (b) Solve the equation $\sqrt{x^2} + 7 = 7 + x$

SECTION B (40 marks)

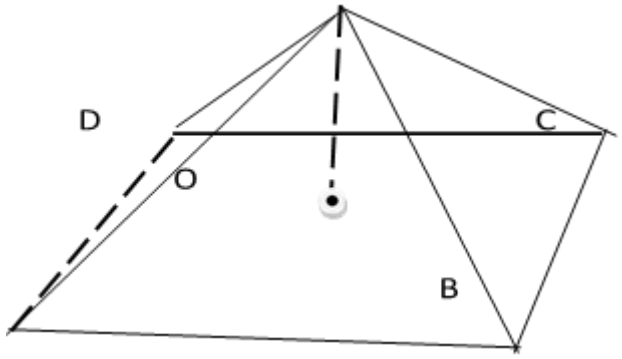
Answer any ALL questions from this section

11. The following marks were obtained by 32 students in a Mathematics examination.

32	35	42	50	46	29	39	64
37	48	52	37	58	52	48	45
37	42	64	37	34	28	58	54
57	54	62	48	67	38	36	34

    i. Prepare a frequency distribution table using the class intervals 24 – 29, 30 – 35 etc
    ii. Draw the histogram
    iii. Draw a cumulative frequency curve and use it to estimate the median.
    iv. Find the mean mark

12. In figure below ABCD is a rectangle in which AB = 3cm and BC = 2cm. V is a point such that VA = VB = VC = VD = 6cm and AO = OC. Find



    a) The angle VAD
    b) The length of AC
    c) The angle between VA and plane

13. (a) Given that $T = \begin{bmatrix} 2-x & x \\ 3 & 2+x \end{bmatrix}$
    i. Write down and simplify the determinant of T.
    ii. Find the value of x for which T has no inverse.
    iii. Determine the inverse of the given matrix and indicate if it is singular or non-singular $A = \begin{bmatrix} 4 & 4 \\ -4 & 4 \end{bmatrix}$

14. (a) Find the image if the line $3x + 4y + 6 = 0$ under a translation by the vector $(-6, -1)$

    (b) A craftsman wishes to decide how many of each type A and B charcoal stove he has to fabricate in order to maximize profit for this month. Unit profit for type A stove is sh.1,000 and Unit profit for type B stove is sh.1,500. Type A stove requires 1m2 of mild steel sheet per unit and type B requires 2m2. He has only 12m2 of mild steel available. He can fabricate at least 8 stoves per month. How many of each type should be fabricated?